

Milestone 1 Documentation: Model Exploration and Technique Development

Project Title: Developing an AI System for Personalized Content
Creation in Media

Intern Name: Naina Anil Pandey

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Milestone 1 Period: February 2 – February 12, 2026

1. Introduction

The "AI-Driven Personalized Content Creation" project aims to build a robust platform that leverages advanced Large Language Models (LLMs) to generate high-quality, tailored content based on specific user commands. The system is designed to produce polished content for LinkedIn posts, advertisements, professional emails, and conversational drafts. The final outcome will be a user-centric intelligent tool hosted on a Streamlit-based web application.

Milestone 1 focuses on exploring and integrating multiple Large Language Models (LLMs) for building a backend system capable of generating personalized textual content.

The objective was to:

- Test different LLM APIs
- Compare their performance
- Handle authentication and API integration
- Understand free-tier limitations
- Build a modular backend for model switching
- Present research findings in a group PPT

2. Milestone 1 Tasks and Objectives

- **Model Exploration:** Researching and testing various LLMs to determine their suitability for diverse content generation tasks.
- **Technique Development:** Studying and implementing prompting techniques to customize LLM responses based on user-defined roles and styles.
- **Content Type Analysis:** Refining the understanding of different formats like ads and social media posts to optimize model outputs.
- **System Reliability:** Identifying multiple LLMs to be used in the project to avoid potential downtime.

3. Tools & Technologies Used

Category	Tools Used
Programming Language	Python
Backend Framework	Flask / FastAPI (as per app.py)
API Testing	Postman
LLM APIs Tested	OpenAI, Gemini, Groq, HuggingFace, Ollama
Environment Handling	.env file
Version Control	Git
OS	Windows

4. Technical Implementation and Testing

I developed several Python scripts using **LangChain** and **Inference APIs** to test model performance.

- **Models Successfully Integrated:**

- **Google Gemini (gemini-2.5-flash):** Used as a high-performance model for dynamic content generation. Fast response and good content quality.
- **Groq (llama-3.3-70b-versatile):** Tested for rapid inference and versatile response generation. Good for experimentation.
- **Ollama (llama3.2:1b):** Implemented for local processing to ensure functionality without external API reliance. No API cost. Works offline once model is downloaded.
- **Hugging Face (zephyr-7b-beta):** Explored as an open-source alternative for simpler explanation tasks. Free tier is limited & the response time is slow but usable.

- **Challenges and Errors:**

- **OpenAI:** While the code for OpenAI (gpt-4o-mini) was prepared, testing was restricted as it requires a paid subscription. I have documented the specific error messages received during these attempts. Error Type: Quota exceeded / Billing not enabled
- **Claude (Anthropic):** Free tier limited. Requires credit card activation.
- **Perplexity:** Requires paid activation for API access. Web version works but API is paid.

5. Group Presentation Summary (Day 5)

Our team presented an "Introduction to Large Language Models (LLM)" covering several critical foundational topics:

- **Definition:** LLMs are sophisticated systems trained on vast datasets to understand and generate human language fluently.
- **Creation Process:** Detailed the stages of Data Collection, Training, Fine-tuning, and Deployment.
- **Core Mechanism:** Explained that LLMs predict language patterns and probabilities using Transformer architecture rather than searching a static database.
- **Generative vs. Static:** Highlighted the shift from simply retrieving existing data to synthesizing unique content for every user.
- **Semantic Understanding:** Demonstrated how LLMs operate on vector representations of "meaning" rather than just manipulating sentences.

6. Free Tier Comparison Table (Models Used)

Platform	Model Used	Free Tier Available	API Free Access	Requires Billing?	Notes
OpenAI	GPT-3.5 / GPT-4	No free credits now	Not sure	Yes	Code works but quota error. Requires credit to use via API.
Gemini	gemini-2.5-flash	Yes	Yes	Not sure	Best free API.
Hugging Face	zephyr-7b-beta	Yes	Yes	No	Free via Inference API (requires an API token).
Groq	llama-3.3-70b	Yes	Yes	No	Free API access for developers (within rate limits).
Ollama	llama3.2:1b	Yes (Local)	Local Only	No	Entirely free (Open Source, runs locally).

7. Student Diary Log: Milestone 1 (Day-wise)

Day	Date	Description
Day 1	Feb 2, 2026	Kick-off Call: Introductory session where interns shared interests, backgrounds, and past experiences.
Day 2	Feb 3,2026	AI Tools & Project Intro: Chat on AI tools and startup stories; project title and query discussed descriptively.
Day 3	Feb 4, 2026	Roadmap & Expectations: Discussed modules, milestones, and flowcharts. Mentors set professional corporate expectations.
Day 4	Feb 5, 2026	Content Exploration: Began documenting different content types and testing prompts against various models.
Day 5	Feb 6, 2026	Group Presentations: Presented our team's PPT on LLM basics; attended peer presentations on RAG and its examples.
Day 6	Feb 9, 2026	Technical Updates: Shared updates on explored LLMs; discussed redundancy strategy using multiple models to avoid downtime.
Day 7	Feb 10, 2026	Priority Discussion: Updates on project progress and ranking LLM usage (Gemini, ChatGPT, and Claude as high priority).
Day 8	Feb 11, 2026	Setup & Limits: Discussed technical setups and model limits; began final submission of Milestone 1 task documentations.
Day 9	Feb 12, 2026	Milestone Finalization: No formal call; focused on individual documentation and finalizing technical evidence for Milestone 1.

8. Milestone 1 Achievements

- ✓ Explored 6+ LLM platforms
- ✓ Understood API integration
- ✓ Handled authentication errors
- ✓ Compared free vs paid models
- ✓ Built modular backend
- ✓ Delivered group presentation
- ✓ Documented working screenshots